

Novel Differentiated Service Methodology based on Constrained Allocation of Resources for Transparent WDM Backbone Networks

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Abstract. *Current Internet traffic comprises diverse service requirements. Service providers differentiate connections characteristics according to demand's requirements and pricing policy. In this paper we propose a differentiated service methodology that implements constrained resources allocation according to demand's priority level for WDM networks. With our proposed mechanism higher priority connections are guaranteed a larger amount of resources, still, lower priority demands are guaranteed a minimum level of service.*

Resumo. *O tráfego da Internet atual possui diversas variedades de requerimento por serviços. Provedores de serviço diferenciam as características das conexões de acordo com as requisições das demandas e a política de cobrança pelo serviço. Neste artigo propomos uma metodologia de diferenciação de serviço que implementa a alocação de recursos constringida de acordo com o nível de prioridade de serviço da demanda em redes WDM. Com o nosso mecanismo proposto conexões com prioridade mais alta são garantidas uma quantidade maior de recursos, mas, demandas de prioridade mais baixa são garantidas um nível mínimo de serviço.*

1. Introduction

Backbone networks offer long-haul data transport to their clients, Internet Service Providers (ISPs). This is a crucial task as a massive amount of data communication is currently provided and Internet communication rates demands are constantly increasing.

Wavelength Division Multiplexing (WDM) is the technology most commonly implemented by Backbone networks. In tandem with a control plane based on the Generalized Multi-Protocol Label Switching (GMPLS) architecture [Mannie 2004], WDM

enables the deployment of transparent networks. A transparent network can currently provide channels with capacity of up to 100 Gbps [Pedrola et al. 2012].

In a WDM network a channel corresponds to a wavelength and an end-to-end path and is known as a lightpath. When a grooming technique is implemented on a WDM network, two or more connections share a wavelength's resources. Grooming techniques are performed electronically and therefore, in a transparent network, traffic grooming is solely executed by the end nodes of a route and connections between a source and destination pair share resources from the same lightpath. A grooming technique may be implemented without following any specific policy or it can follow a given strategy that may regard Quality of Service, differentiated service and/or protection strategies.

When network's resources are not enough to serve all current clients' requests some demands will be blocked. When different level of services are established in the network, network managers may follow a policy to determine to which demands prioritize service. Minimum service level guarantees are specified by Service Level Agreements (SLA) signed between the data transport provider and its various clients. By implementing a differentiated service provisioning strategy data transport providers may obtain a cost effective service provisioning by avoiding SLAs penalties and optimizing network resources use [Song et al. 2007].

In order to offer different service levels, backbone networks implement differentiated service mechanisms. In WDM networks the most common service mechanisms assign a path based on its availability profile or providing different protection and/or restoration schemes according to a demand's priority of service.

Traffic pattern is usually unknown in advance, for this reason preemption procedures are necessary to avail resources to higher priority connection demands that may arrive during network operation. In a virtual circuit packet based network, as Multi Protocol Label Switched (MPLS), a traditional and commonly differentiated service approach is the preemption of lower priority connections to accommodate arriving higher priority traffic as described in [Le Faucheur et al. 2003] and [de Oliveira et al. 2007]. In a network where preemption is enabled, a higher priority traffic may use resources originally assigned to a lower priority traffic by disruption of its Label Switched Path (LSP).

Following SLA guidelines it is the network manager's responsibility to classify a network's traffic into priority levels. In MPLS networks the priority levels determine the priority of a demand to request preemption of an LSP. A backbone network connection transports a large volume of data and therefore should, ideally, not be disrupted. It is for this main reason that preemption strategies are not strictly adequate to backbone networks. Furthermore, preemption policies tend to cause instability, excessive number of re-routing decisions [de Oliveira et al. 2004] and a large number of Resource Reservation Protocol (RSVP) messages exchange each time an LSP is preempted [Salvadori and Battisti 2004]. All the aforementioned characteristics have a negative impact on large and complex networks as backbone networks.

Different preemption methodologies have been proposed with the objective of minimizing instability and message exchange overhead, proposals may be observed in Section 2. A differentiated distribution of resources approach is described for MPLS architecture in [Le Faucheur et al. 2003]. In DiffServ-TE MPLS LSPs are established with

a given bandwidth guarantee. Connections are organised into classes of traffic defined by a class-type (CT). The amount of bandwidth reserved varies according to a connection's CT value.

We propose a mechanism to implement differentiated resources distribution in WDM networks for long-haul data transport. With our proposed mechanism backbone network managers may design different profiles of service for the network's traffic, and distribute the network's resources aiming at financial optimization. We tested the proposed mechanism in an ad-hoc simulator and results indicate that by implementing our proposed mechanism higher priority connections have their service benefited, therefore optimizing network's revenue/cost ratio. In addition, lower priority connections are not disrupted and are guaranteed a minimum amount of allocated resources.

The rest of this paper is organized as follows. Section 2 presents related work on the topic. Section 3 describes the proposed mechanism. Section 4 explains the performance study and presents the results and their analysis. Finally, Section 5 concludes and introduces future works.

2. Related Work

Both GMPLS and MPLS networks are circuit switched networks where a connection's resources requirement is reserved before data transport begins. Preemption or differentiated distribution of resources policies may be implemented in MPLS as well as in GMPLS networks. In our work we extend a mechanism originally designed for MPLS networks to GMPLS networks. Therefore we review available mechanisms for MPLS networks as they may be adapted for GMPLS networks.

Authors in [Meyer and Vasseur 2010] propose a preemption scheme in which a connection selected to be preempted is re-routed and therefore not disrupted. In their proposal the ingress Label Edge Router (LER) under alert may trigger the necessary process to enable a temporary under-provisioning. Similarly, in [de Oliveira et al. 2004], an adaptive scheme is proposed for MPLS to select lower priority LSPs that can afford having their rate reduced in order to satisfy a higher priority connection requirement. The mechanism we propose is also based on under provisioning of lower priority connections, adapted for GMPLS WDM networks.

In [Sone et al. 2011] authors propose temporary under provisioning as a recovery scheme in Spectrum Switched Optical Networks (SSON). In case of a failure a low priority connection is switched to a backup path and its bandwidth provision is reduced. The proposal does not specify a model for resource reduction. In our proposal we study an specific resource constraint model.

In a WDM network most common differentiated service mechanisms are based on resilience strategy and/or link availability profile. A link's availability profile is determined by the amount of time the link works without failure. Both in [Song et al. 2007] and [Tornatore et al. 2012] authors propose a grooming approach in which connection requests are groomed into a lightpath according to its availability profile. In [El-Gorashi and Elmirghani 2009] authors propose a model with three resilience classes: shared, dedicated protection or restoration. Following this model traffic grooming is performed regarding demands' resilience requirements.

An alternative differentiated service approach was studied in [Dantas et al. 2012]. In this work authors focus on the Path Computation Element (PCE) architecture [Farrel et al. 2006] and analyse the impact of the priority in queue of a path computation request message (PCReq), described in [Vasseur and Roux 2009], on the resulting lower blocking probability.

To the best knowledge of the authors, there is no published work proposing differentiated distribution of resources nor there is any work concerning partial resources preemption for WDM networks. Our proposed mechanism offers an innovative differentiated service mechanism providing tools not currently available for WDM network managers.

3. Proposed Mechanism

In this paper, we propose a new mechanism for WDM networks based on differentiated distribution of resources developed MPLS networks in [Le Faucheur 2005b]. In the original MPLS mechanism a CT may be understood as a set of traffic trunks crossing a link and on which a specific set of bandwidth constraints is enforced [Tornatore et al. 2012].

When a differentiated distribution of resources mechanism is implemented in a network the resources assignment process must follow the constraints determined by a bandwidth constraint model. Three models were proposed for MPLS networks. In Maximum Allocation Model [Le Faucheur 2005a] (MAM) each CT has a maximum amount of resources that can be assigned to it as an upper limit. Maximum Allocation with Reservation (MAR) model is similar to MAM but the upper limit of bandwidth can be disregarded if the link being considered is not above a certain level of usage. In the Russian Dolls Model (RDM) [Le Faucheur 2005c] each class shares its maximum amount of bandwidth assignment with higher CTs. The lowest CT value's maximum bandwidth assignment equals the link's capacity and resources that are not being used by a higher CT may be used by lower CT connections.

Each resources constraint model has its benefits and disadvantages depending on networks management needs and network's characteristics. Still, there are performance analysis in the literature that point to advantages on implementing RDM. The RDM mechanism presented better performance regarding low priority connections, [Goldberg et al. 2006] and higher levels of connections' QoS [Molnar and Vlcek 2009]. RDM was proven to best adapt to the Diff-Serv bandwidth allocation policy by achieving high link utilization and guaranteeing quality of service to the various Diff-Serv classes [Lai 2005]. For the reasons aforementioned we have selected the RDM as the model used by our proposed mechanism.

In our proposed mechanism the resources constrained distribution idea is extended to a WDM network. In the WDM network scenario, we consider in our study, we assume path computation and resource assignment would be performed centrally by a PCE as illustrated by Figure 1. In the figure it is possible to observe a PCE exchanging messages with GMPLS nodes that work as Path Computation Clients (PCC). PCCs require path and resource computation to the PCE for a given demand (or demands) at hand. After a PCE's successful reply message it is up to the GMPLS architecture protocols to reserve and assign lightpaths to connections. In order to be able to implement the RDM algorithm a domain's PCE would have to be enhanced.

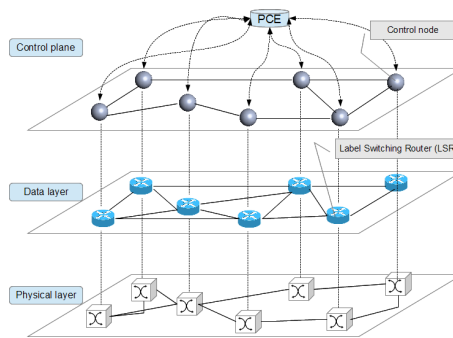


Figure 1. A WDM network with path computation centralized by PCE with communication, via PCE protocol, with control nodes

In the Russian Dolls Model, a link's maximum capacity use is achieved by accumulating successive CTs and bandwidth constraints (BC). For example CT7 may use resources up to BC7 limit, while CT6 shares its BC6 constraint with CT7, CT5 shares its BC5 constraint with CT6 and CT7 and so on as illustrated by Figure 2. Therefore, a lower priority traffic may use a higher priority class's bandwidth reservation, in case it is not in use, up to its own bandwidth constraint value, while a higher priority connection can preempt a lower priority connection up to its own bandwidth constraint value [Le Faucheur 2005c].

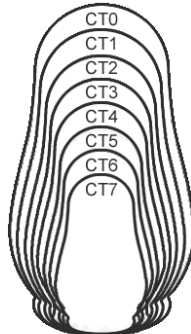


Figure 2. RDM Russian Dolls Model

In our proposed mechanism resources preemption is performed on a first fit basis and, to prevent excessive connection disruption and resource re-assignment, when a higher priority connection is torn down its resources are not re-assigned to lower priority connections.

The bandwidth constraints for each class must be determined by network management taking into account SLA details. The BC values must be carefully determined as they define the level of privilege over resources that higher priority connections will have. For instance, in a situation where one channel is assigned connections with all possible CT values and the difference between each BC value is equal for most CT values, different CT connections could end up having the same amount of resources assigned. However, if the difference between BC values increases as the CT value increases higher priority connections would be guaranteed more resources than lower priority ones.

The RDM is a resource distribution model. Throughout this paper, however, we will refer as RDM mechanism as the mechanism that implements this resource distribution model when performing the RWA algorithm.

3.1. Algorithm Description

In our proposal, the RDM mechanism is triggered when contention for resources occur, which will be detected in the network when a connection request is blocked by the grooming mechanism. When triggered, the RDM algorithm implements a resource distribution constraint model so when a demand arrives the lightpath computation process will limit the bandwidth requirement to its respective BC value. We also tested a variation of the mechanism where the assignment of resources follows no constraint. In this mechanism version the BC values are used to determined the constrained amount of resources that may be preempted from the various CT value connections. We will call this version of the mechanism RDM Aggressive herein.

In the algorithm, path computation is performed offline as alternate fixed routes. Having established the amount of resources to be assigned, the algorithm selects the first route from 3 alternate fixed possible routes. Then, in first fit order, it searches for a lightpath containing a connection using the same route as the one selected, so that grooming may be performed. If such a lightpath is found, the algorithm checks if the lightpath's available resources are enough to accommodate the demand at hand. In case it does, the demand is assigned as a new connection to the lightpath.

If there are not enough resources available, the lightpath's id is recorded in case the RDM mechanism needs to be triggered later. If, however, a lightpath containing a connection using the selected route is not found, the algorithm searches, in first fit order, an empty lightpath in the selected route. If the process is not successful the tasks described above are consecutively performed for 2nd best and then 3rd best routes.

If a connection cannot be established after the previous steps, but a lightpath containing a connection using the same route was found, then the RDM mechanism is triggered. If the version adopted is not RDM Aggressive, at this moment, the bandwidth constraint is enforced on demand's BW requirement. By accessing the information outputted from the grooming process the RDM algorithm already knows the lightpath supports data to be groomed into it. In the selected lightpath, it searches for connections with the lowest possible priority value and calculates the amount of resources that can be preempted from the selected connection. In case the resulting re-usable resource is not enough the procedure is repeated for all connections with the same lower CT value. This process is repeated for the consecutive CT values as long as their CT value is not equal or higher than the demands and until the demand's BW requirement is met by the sum of all preemptable resources.

If after checking all the lightpath's connections the resulting total amount of resources that can be preempted is not enough to satisfy the demand's request then the demand is blocked. Conversely, if enough resources are found the lightpath is selected by the PCE as the resulting computed path. For the demand to be assigned as a connection to the lightpath, the suggested lightpath and possible reduced suggested resources must be accepted, in which case, the channel's current connections are updated.

Table 1. BW requested values distribution

BW requested value	10	40	100
Probability	0.80	0.15	0.05

Table 2. CT values distribution

CT value	0	1	2	3
Probability	0.60	0.25	0.10	0.05

4. Performance Study and Results Analysis

We developed a Java based discrete event driven simulator in order to evaluate the performance of our differentiated resources distribution algorithm. We performed a simulation using the German Nobel reference network topology with 17 nodes and 26 bi-directional links [Orlowski et al. 2010]. We consider that each link in the network corresponds to one fibre comprising wavelengths with 100 Gbps bandwidth capacity each [Pedrola et al. 2012].

Connection requests are generated dynamically following a uniform distribution among node pairs for source and destination. All runs are averaged over 5×10^5 requests following a Poisson process with mean holding time (λ) set to 500 sec (exponential). Request inter-arrival time (μ) are also exponential and vary with loading. Offered traffic load is calculated in Erlang using the relation $E = \lambda \mu$ [Ghimire and Mohan 2012]. Bandwidth requests are non-uniformly distributed among 10, 40 and 100 Gbps, with distribution approximate to values from [Pedrola et al. 2012] that can be observed in Table 1.

Previously, we tested the algorithm using different numbers of CT values. We observed that 4 CT values represent a good trade-off between complexity and overall performance. Therefore we decided to determine 4 possible CT values for the demands in the simulation, where 0 is the lowest priority value and 3 is the highest priority value. The 4 CT values follow a non-uniform distribution that can be observed in Table 2.

We evaluated the algorithm for different BC models but, due to space limitations, we selected to present results for the model that presented a higher level of benefit. The selected model's values are expressed in Table 3. BC values refer to the maximum amount of resources allowed to be allocated to a demand, minimum BW guaranteed is the amount of resources that will not be preempted from an active connection with a given CT value and Max. robbed is the maximum amount of resources a demand with a given CT value

Table 3. BC Model values

CT value	0	1	2	3
BC values (in Gbps)	100	97	79	47
Minimum BW guaranteed (in Gbps)	3	18	32	47
Max. robbed (in Gbps)	0	3	21	53

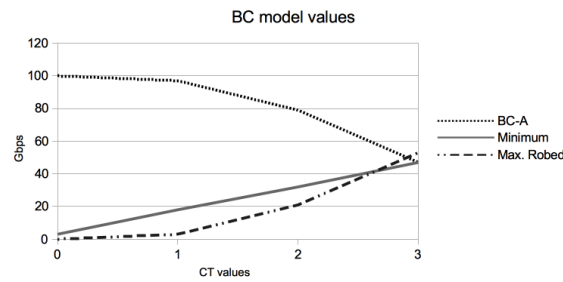


Figure 3. Constraint values for BC model A

may use from resources preempted from connections with lower CT values. Figure 3 illustrates the relation between the BC values and each of the CT values.

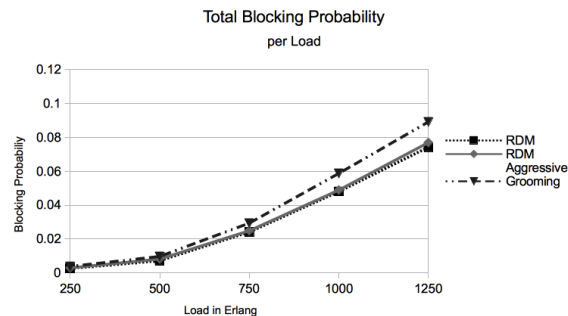


Figure 4. Blocking probability for RDMs and grooming scenarios for various network offered loads

Performance metrics analysed in the simulations were blocking probability and amount of resources allocated to connections. Simulation results reached a 98% confidence interval level. We compared the performance of the proposed mechanism's two versions, RDM and RDM Aggressive, with the resulting performance of a network scenario where a grooming mechanism is implemented without connection assignment policy.

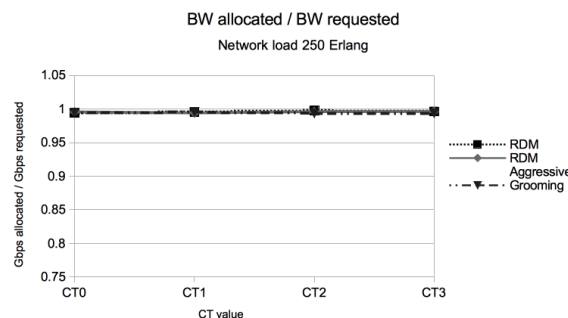


Figure 5. Blocking probability by CT values for RDMs and grooming scenarios for a network offered load of 250 Erlang

Simulation results indicate that as network offered load increases the RDM mechanism is triggered more often and the blocking probability tends to be increasingly smaller than the blocking probability of the grooming scenario as can be observed in Figure 4. In

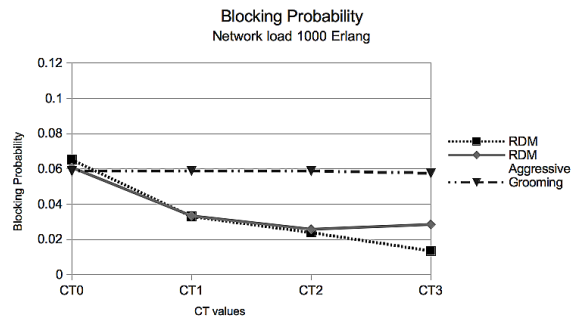


Figure 6. Blocking probability by CT values for RDMs and grooming scenarios for a network offered load of 1000 Erlang

our experiment, when the network offered load is 250 Erlang the blocking probability for all the three scenarios analysed may be considered to be the same, that is justified by the fact that under light offered load contention for resources do not occur often and the RDM mechanism is thus seldom triggered. In a network scenario with a heavy load the difference in blocking probability can be of 1% for a network offered load of 1000 Erlang and up to 2% for a network load of 1250 Erlang as can be observed in Figure 4. It is also possible to observe that, when the total number of demands is being considered, both RDM mechanism versions implemented provide approximately the same decrease in blocking probability when compared to the grooming scenario.

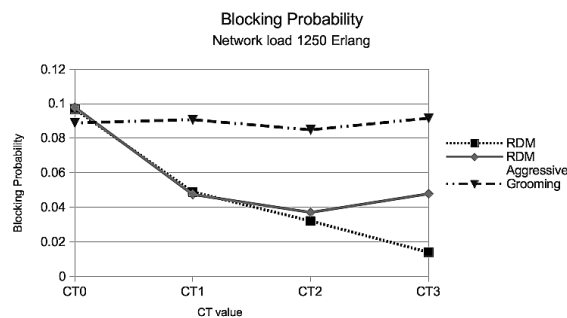


Figure 7. Blocking probability by CT values for RDMs and grooming scenarios for a network offered load of 1250 Erlang

When the blocking probability is analysed according to the demands' different CT values, in the scenario where the network offered load was 250 Erlang (Figure 5) the blocking probability is very low for all demands with different CT values, and therefore difference in the blocking probability values is unnoticeable. However, when the network offered load is 1000 Erlang or 1250 Erlang the blocking probability decreases as the connections' CT values increase. As can be observed in Figure 6 and in Figure 7 the largest beneficial impact on blocking probability performance occurred for connections with the highest priority, the ones belonging to CT value 3. In the RMD scenario the blocking probability for demands belonging to CT value 3 decreased almost 5% if compared to the grooming scenario for the scenario with an offered load of 1000 Erlang (Figure 6) and a decrease of almost 8% for an offered load of 1250 Erlang (Figure 7). In the RDM Aggressive scenario, however, the level of blocking probability decrease was less pronounced, being around 3% decrease for 1000 Erlang (Figure 6) and less than 5% decrease for the

1250 Erlang (Figure 7) offered load scenarios.

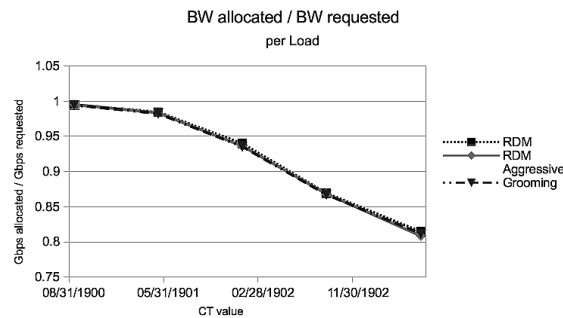


Figure 8. Proportional resources allocation for RDMs and grooming scenarios according to different network offered load

This result shows that the RDM Aggressive mechanism version does not offer best results for highest CT value connections. This phenomenon occurs because, in the RDM Aggressive version, each CT3 connection is allocated a larger amount of resources, and therefore, is responsible for a larger use of channels resources than in the RDM scenario. CT3 demands cannot preempt resources from other CT3 connections, therefore tend to compete between each other. In a network where a larger amount of resources are being used by CT3 connections, CT3 demands will have a higher blocking probability as they will have less resources to preempt for their own use. CT2 and lower CT values connections were also granted their original BW requirement, however due to the fact their allocated resources may be preempted by a higher CT value demand the resulting amount of resources allocated to these connections are limited. The fact that lower CT value connections have a limit on their resources allocation represents less competition for resources between connections of the same lower CT value.

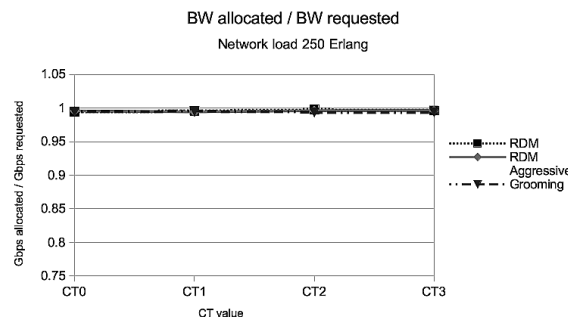


Figure 9. Proportional resources allocation for RDMs and grooming scenarios for different CT value connections with a network offered load of 250 Erlang.

We have also analysed the proportion of resources allocated to connections in relation to the amount of resources requested by demands. When all the connections were considered together it was possible to observe that implementing any of the two versions of the RDM mechanism the amount of proportional resources allocation did not increase in any of the network offered load scenarios as illustrated by Figure 8. In order to avoid repetition of terms, in the rest of this paper, we refer to proportional resources allocation simply as resources allocation.

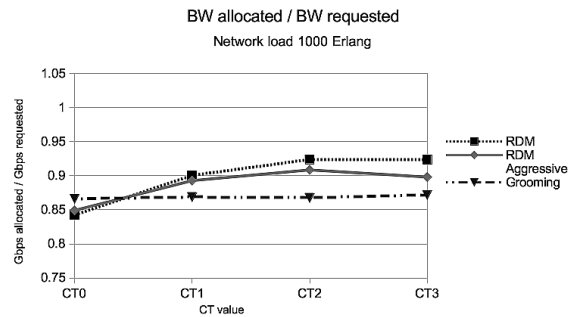


Figure 10. Proportional resources allocation for RDMs and grooming scenarios for different CT value connections with a network offered load of 1000 Erlang.

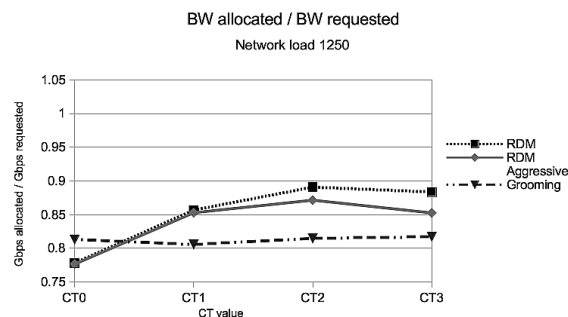


Figure 11. Proportional resources allocation for RDMs and grooming scenarios for different CT value connections with a network offered load of 1250 Erlang.

When the allocation of resources is analysed according to connections' CT values, the difference in the BW allocation between the different scenarios is only noticeable for network offered load of 500 Erlang or higher. Indeed with a network offered load of 250 Erlang it is possible to affirm there is no difference in resources allocation and distribution between the different scenarios as can be observed in Figure 9.

For higher network offered loads, however, the RDM mechanism tends to allocate more resources to higher CT value connections than the RDM Aggressive mechanism. For a scenario where the network offered load is 1000 Erlang, as illustrated by Figure 10, the resources allocated to connections with CT values 3 and 2 have an increase of approximately 5% while for the same network offered load the RDM Aggressive mechanism resulted in approximately 3% increase for connections with CT values 3 and 2. For a scenario where the network offered load is of 1250 Erlang, as illustrated by Figure 11, by implementing the RDM mechanism the proportional resources allocated to connections with CT3 and CT2 increase in approximately 7% while for the same network offered load the RDM Aggressive mechanism resulted in approximately 4% increase for connections with CT values 3 and 2.

This phenomenon can be explained by the fact that in RDM Aggressive mechanism version less connections of CT3 are being established, but at the same time they require a larger amount of resources than in RDM mechanism scenario. These larger amount of resources required from higher CT value demands will be preempted from the lower CT value connections, therefore CT2 connections also present a decrease in the resources allocated when compared to the RDM mechanism scenario.

As for the lowest CT value connections both versions of the RDM mechanism decrease the resources allocated for those connections when compared to the grooming scenario. This fact explains why the resulting resources allocated to all connections do not differ between different values of offered load: for all scenarios analysed the total amount of resources allocated is virtually the same, but the allocation is distributed differently to the different CT value connections.

Blocking probability and resources allocation results have a direct impact on a network's revenue. The service fee of higher priority connections is usually higher than the fee of lower priority connections. A pricing policy may be based on the number of connections established or the amount of resources allocated to a connection. In both cases it is possible to infer that by increasing service level to higher priority connections increases the amount of collected service fee. As an example, [Xia et al. 2009] presents service fee values for four different classes of service. We calculate the proportional service fee values according to a service fee value of reference, in this case the value of service fee for CT1, which results in the following proportional fee according to CT value : CT0 0.65; CT1 1.00; CT2 1.50; CT3 2.00. From these values it is possible to observe that offering service to a connection with CT3 generates twice as much collected fee as offering service to a connection with CT1.

In this section we demonstrated that the proposed RDM mechanisms perform differently from the grooming mechanism in terms of blocking probability and resources distribution according to connections' CT values. As the network's offered load increases the RDM mechanism is triggered more often. As a higher number of connections are established via the RDM mechanism higher priority connections benefit from a smaller blocking probability and larger amount of allocated resources. Since higher priority connections tend to have stricter SLA requirements, and are usually charged more by SPs, increasing the service for higher priority connections will eventually increase a SP's revenue.

5. Conclusion and Future Work

We proposed a novel mechanism for differentiation of service for a WDM network based on the constrained distribution of resources. The mechanism implements a resources distribution constrained model called the Russian Dolls Model (RDM). The RDM determines resources distribution and allocations constraints according to different levels and priorities of service. We tested two versions of the mechanism in an event driven simulation and analysed their resulting performance in comparison to a scenario where a grooming technique is implemented without a resource distribution policy.

From simulation results we observed that our proposed mechanism increases the distribution of resources to higher priority connections. Furthermore, the proposed mechanism decreases overall blocking probability for heavy load traffic scenarios. We believe that, the implementation of the mechanism, in a network, promotes better service to higher priority connections resulting on a net revenue increase. This economic benefit is justified by the fact that a SP's both service fees and penalty fees are higher for higher priority connections.

We are aware that in order for the mechanism to be properly implemented on a network control plane enhancements are required. Specifically, extensions on PCE pro-

ocol and GMPLS' RSVP-TE are necessary to guarantee the mechanism's deployment. However this subject is out of the scope of this article and will be addressed by future works.

As future works we will extend the study of differentiated service mechanism proposals for Elastic Optical Networks. And we test the proposed mechanisms in different topologies.

6. References

References

- Dantas, J., Careglio, D., Silveira, R., Ruggiero, W., and Sole-Pareta, J. (2012). PCE QoS tools and related scalability in WDM networks. In *Transparent Optical Networks (ICTON), 2012 14th International Conference on*, pages 1–7. IEEE.
- de Oliveira, J., Vasseur, J., Chen, L., and Scoglio, C. (2007). Label switched path (LSP) preemption policies for MPLS traffic engineering. Technical report, IETF RFC 4655, August.
- de Oliveira, J. C., Scoglio, C., Akyildiz, I. F., and Uhl, G. (2004). New preemption policies for DiffServ-aware traffic engineering to minimize rerouting in MPLS networks. *IEEE/ACM Transactions on Networking (TON)*, 12(4):733–745.
- El-Gorashi, T. E. and Elmirghani, J. M. (2009). Differentiated resilience with dynamic traffic grooming for WDM mesh networks. In *Transparent Optical Networks, 2009. ICTON'09. 11th International Conference on*, pages 1–6. IEEE.
- Farrel, A., Vasseur, J.-P., and Ash, J. (2006). A path computation element (PCE)-based architecture. Technical report, IETF RFC 4655, August.
- Ghimire, R. and Mohan, S. (2012). A token-based routing mechanism for GMPLS-controlled WDM networks. *Optical Switching and Networking*, 9(2):170–178.
- Goldberg, J. B., Dasgupta, S., and de Oliveira, J. C. (2006). Bandwidth Constraint Models: A Performance Study with Preemption on Link Failures. In *GLOBECOM*.
- Lai, W. S. (2005). Bandwidth constraints models for differentiated services (DiffServ)-aware MPLS traffic engineering: performance evaluation. Technical report, IETF RFC 4128, June.
- Le Faucheur, F. (2005a). Maximum allocation bandwidth constraints model for DiffServ-aware MPLS traffic engineering. Technical report, IETF RFC 4125, June.
- Le Faucheur, F. (2005b). Protocol extensions for support of Diffserv-aware MPLS traffic engineering. Technical report, IETF RFC 4124, June.
- Le Faucheur, F. (2005c). Russian dolls bandwidth constraints model for DiffServ-aware MPLS traffic engineering. Technical report, IETF RFC 4127, June.
- Le Faucheur, F., Lai, W., et al. (2003). Requirements for support of differentiated services-aware MPLS traffic engineering. Technical report, IETF RFC 3564, July.
- Mannie, E. (2004). Generalized multi-protocol label switching (GMPLS) architecture. Technical report, IETF RFC 3945, October.

- Meyer, M. and Vasseur, J. (2010). MPLS traffic engineering soft preemption. Technical report, IETF RFC 5712, January.
- Molnar, K. and Vlcek, M. (2009). Evaluation of bandwidth constraint models for MPLS networks. *J Electronic*, pages 1313–1842.
- Orlowski, S., Wessály, R., Pióro, M., and Tomaszewski, A. (2010). SNDlib 1.0 Survivable network design library. *Networks*, 55(3):276–286.
- Pedrola, O., Castro, A., Velasco, L., Ruiz, M., Fernández-Palacios, J., and Careglio, D. (2012). CAPEX Study for a multilayer IP/MPLS-over-flexgrid optical network. *Journal of Optical Communications and Networking*, 4(8):639–650.
- Salvadori, E. and Battisti, R. (2004). Quality of service in IP over WDM: considering both service differentiation and transmission quality. In *Communications, 2004 IEEE International Conference on*, volume 3, pages 1836–1840. IEEE.
- Sone, Y., Watanabe, A., Imajuku, W., Tsukishima, Y., Kozicki, B., Takara, H., and Jinno, M. (2011). Bandwidth squeezed restoration in spectrum-sliced elastic optical path networks (SLICE). *Journal of Optical Communications and Networking*, 3(3):223–233.
- Song, L., Zhang, J., and Mukherjee, B. (2007). Dynamic provisioning with availability guarantee for differentiated services in survivable mesh networks. *Selected Areas in Communications, IEEE Journal on*, 25(3):35–43.
- Tornatore, M., Lucerna, D., Mukherjee, B., and Pattavina, A. (2012). Multilayer protection with availability guarantees in optical WDM networks. *Journal of Network and Systems Management*, 20(1):34–55.
- Vasseur, J. and Roux, J. (2009). Path computation element (PCE) communication protocol (PCEP). Technical report, IETF RFC 5440, March.
- Xia, M., Tornatore, M., Martel, C. U., and Mukherjee, B. (2009). Service-centric provisioning in WDM backbone networks for the future internet. *Journal of Lightwave Technology*, 27(12):1856–1865.